

STRATEGIC MEDIATION OF INFORMATION IN AUTOCRACIES

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Abstract

This paper presents the optimal editorial policy for the state-owned media manipulating information flow from a strategic informed elite to an uninformed receiver. The receiver attempts to match the state of the ruler's competence drawn from the unit interval with a binary action. If the elite's and audience's preferences are too distant from each other, then the editorial policy is uninformative. Otherwise, the media signal whether the state is higher or lower than a threshold which depends on the elite's preferences. An elite that is more aligned with the media benefit the media, as long as the alignment is not too close. The media are worse off when the receiver is more critical of the ruler, whereas the elite generally is better off when the receiver is more critical. When the receiver has private information about how critical he is, I characterize the lower bound on the media's payoff obtained within the class of restricted editorial policies. I identify a sufficient condition that implies the bound is achieved.

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1 Introduction

The dominant model of dictatorship has evolved over the course of the twentieth century. The authoritarian states rely less on terror and ideology to make the citizens abide by ruler's political objectives than before. Softer autocracies have emerged, including Russia, Venezuela, Ecuador, Turkey before the coup attempt in 2016, among others. These states no longer practice massive repressions. Instead, they hold elections and allow legal opposition in the attempts to imitate democracy (Gandhi and Lust-Okar, 2009). As claimed by Guriev and Treisman (2019), these states seek to convince the population in ruler's competence to lead the country into a prosperous future. One of the main instruments for achieving this goal is an information manipulation through multiple channels including state-owned media. That is why such authoritarian states are sometimes referred to as informational autocrats.¹ The state-owned media consistently manipulate facts and censor information to influence citizens' beliefs about the ruler's competence. These beliefs then play a role when the population is faced with a decision whether to adhere to the ruler's political target.

In many situations, the state-owned media do not have access to the facts. Instead, they have to rely on reports generated by a strategically-interested information source. For example, reports could be research conducted by an independent statistical agency in hopes of providing the most accurate information to the general public. Reports could also surface from ruler's cabinet members acting in their own best interest. In the presence of a strategic source, how likely is swaying citizens' decisions toward the ruler's favor? What is the optimal editorial policy for the state-owned media and how does the policy depend on the preferences of citizens and the source?

To answer these questions, I consider a model of the optimal information disclosure by a state-owned media to a representative uninformed receiver. The receiver must decide between taking a mobilizing or the status-quo actions. Some examples of mobilization include voting for the ruler in the election, voting in favor of the ruler's proposal to change the Constitution, not revolting, not going to anti-government protests (Gehlbach and Sonin (2014), Shadmehr

¹This term is used in Guriev and Treisman (2019) that provides an extensive overview of the inner processes in modern autocracies.

and Bernhardt (2015)). The media want the receiver to take the mobilizing action.² However, the receiver prefers the mobilizing action only if the ruler's competence is high enough. The media do not have access to the ruler's competence and such information has to be supplied by the informed elite. The elite knows the state of the ruler's competence. This knowledge can come from an independent research, a proximity to the ruler, or an ability to understand political processes better than the receiver (Guriev and Treisman, 2018). The elite's ordinal preferences are such that only if the elite observes that the ruler's competence is above a certain threshold, then she prefers the mobilizing action. The elite cannot communicate to the receiver directly. Instead, having learnt the competence, the elite sends a message to the media. The media then generate a report to the receiver. Finally, the receiver chooses an action based on the media's report. I study the media's problem under the commitment assumption. That is, at the beginning of the interaction, the media's editorial policy on how reports are generated from elite's messages is announced.

The media's optimal editorial policy is simple to describe. If the elite and the receiver disagree on the favorable action for a sufficiently large set of the ruler's competence levels, then the media cannot do better than providing no information and the receiver opts for the status-quo action.³ Otherwise, the media signal whether the ruler's competence is higher or lower than a threshold which depends on the elite's preferences. By doing so, the media ensure that the elite reports information truthfully. If the ruler's competence is high enough then the editorial policy suggests the mobilizing action to the receiver. Otherwise, the status-quo action is suggested with some probability. This probability is calibrated to make the receiver indifferent between two actions. I show that if the receiver becomes more critical of the ruler, that is, he requires a higher ruler's competence to oblige with choosing the mobilizing action, then the media are worse off. It becomes harder for the media to convince the receiver to choose the mobilizing action. I also show that the elite generally benefits when the receiver becomes more crit-

²The ruler needs to delegate the responsibility for reporting news to a designated institution. The ruler's interests are presented by the state-owned media with the same objective, even though the ruler does not participate in the game.

³The intuition is similar to the one in sender-receiver games, where if the sender's bias is too large, then the equilibrium is necessarily uninformative.

ical. As discussed in [Guriev and Treisman \(2019\)](#), highly educated citizens in the authoritarian states tend to be more critical toward their government. Thus, as a prediction of this model, the spread of higher education would make it harder for the media to sway the receiver toward the mobilizing action and make the informed elite better off. The media benefit from the elite that is more aligned with the media as long as the alignment is not too close: reports generated from the source having a similar objective to the media's lead to the receiver's skepticism.

The media need to incentivize the elite to supply information by designing the editorial policy tailored to the elite's preferences. Such honesty constraints discipline the editorial policy and render some information sources to be futile to the media. This paper helps to understand the relationships between the informed elites and the state-owned media. In particular, I show that the elite closely aligned to the ruler's objective is fruitless to the state-owned media if the receiver is able to correctly make inferences. Importantly, even if the elite is heavily critical of the ruler, the state-owned media can still weaponize elite's information to persuade the receiver.

The initial analysis assumes that the population is identical in how critical they are, and hence it can be represented by a single representative receiver. However, in reality, the population is heterogeneous in pickiness towards the ruler, even in authoritarian states.⁴ This observation leads me to consider the state-owned media attempting to sway decisions of the population of receivers having private information about their pickiness. Without honesty constraints, the state-owned media implement the upper censorship policy revealing low levels of the ruler's competence and pooling high levels ([Kolotilin et al., 2017](#)). Honesty constraints reduce the media's welfare, since the upper censorship policy cannot be implemented. Revealing all low states of the ruler's competence is not available due to incentives of the low types of the source. I characterize the solution to the media's problem choosing over the restricted simple class of editorial policies with a support having at most two elements. The optimal simple policy provides the lower bound for the media's unconstrained problem. I show the sufficient

⁴For example, Russian independent pollster Levada center reports respondents' answers to the question "Do you approve the decisions of Vladimir Putin as the president of Russia?" in October 2019. 26 % of the respondents answered "Yes, absolutely", 44 % answered "Rather yes, than no", 18 % answered "Rather no, than yes", 10 % answered "Absolutely not", 2 % abstained. URL: <https://www.levada.ru/2019/11/18/vladimir-putin-7/> (available in Russian only). See also [Neundorff et al. \(2020\)](#).

condition on the distribution of receiver's private information, under which this lower bound is attained. I illustrate the lower bound for the unimodal distribution of receiver's types.

2 Related literature

This paper is a part of an active literature that studies *strategic information dissemination decisions by the media* concerned with swaying the beliefs of its audience. Guriev and Treisman (2018) present the model of the informational autocracy that yearns for staying in power. A competent ruler brings higher living standards. As in my model, there is an elite that is informed about the ruler's competence. The elite may send messages to the population. These messages can be censored by the ruler, or the ruler may buy silence off the elite. The ruler is also able to send propaganda directly to the population. The authors establish conditions under which the manipulation of information is more beneficial for the ruler than opting for repressions or improving living standards. In my paper, the only way for the elite to communicate with the public is through the state-owned media. The media do not generate information itself, and transfers are not allowed. The manipulation of information is the only available instrument to the media. I characterize the effectiveness of this instrument depending on the preferences of the elite, the media, and the population.

Most of the papers in the literature assume that news is exogenous in the sense that it is a realization of the payoff-relevant random variable for the audience. Shadmehr and Bernhardt (2015) explore a ruler's decision of whether to censor information available to citizens to avoid a revolution. The citizen's net payoff from successful revolution depends on the news that can be censored by the ruler. The authors characterize the ruler's censorship strategy and show that the ruler is better off by committing to censor less than he does in the equilibrium. Duggan and Martinelli (2011) consider the election model with an incumbent and a challenger in which the media can affect the public opinion. The state of the world is the challenger's policy on a level of public good provision and an income tax rate. As in my model, the media can commit to how it systematically distorts the information about the challenger's policy. The authors characterize the choice of the media slant for pro-incumbent and pro-challenger media. The

slanting technology is fixed and represented by the projection of the two-dimensional policy on a straight line with a slope representing the media slant. Instead, in my model, I consider the general slanting technology for the one-dimensional state and allow the information supplier to be strategic.

Chiang and Knight (2011) and Gehlbach and Sonin (2014) provide empirical evidence that voters take media's bias into account when forming beliefs about political candidates. Gehlbach and Sonin (2014), Duggan and Martinelli (2011), and Gentzkow et al. (2015) adopt the assumption of the media's commitment power to a probabilistic information structure, as my paper does. This assumption captures the government's need to delegate responsibility for reporting news to correspondents, reporters, and editors who make frequent decisions about the framing of the news they decide to cover.

This paper also contributes to the literature on *mediated cheap talk*, which studies communication between an informed sender and an uninformed receiver through the mediator. The informed party makes a report to the mediator, who then makes a non-binding recommendation to the receiver. The literature focuses on the optimal mediation for the sender and the receiver. The optimal mediation generally adds noise to communication. Goltsman et al. (2009) characterize the optimal mediation for the uniform-quadratic setup of the cheap-talk game of Crawford and Sobel (1982). Blume et al. (2007) analyze the special case of the mediation protocol: with some probability, the sender's message is transmitted perfectly to the receiver; with the remaining probability, the noisy message is generated. The authors show that noise generally improves welfare. They also derive the optimal level of noise that achieves the best sender's and receiver's payoffs across all communication devices. These papers concentrate on the neutral mediator and characterize the best mediator for the sender and the receiver. In my paper, the media play the role of the mediator and have a strategic objective, specifically, to increase the probability of the mobilizing action chosen by the receiver. I analyze the optimal mediation plan for different assumptions on the sender's and receiver's preferences. Within the uniform-quadratic setup, Ivanov (2010) shows that there is no welfare loss if the strategic mediator is chosen properly. Compared to this paper, I assume the media's ability to commit to the mediation plan. Therefore, the media in my setup will generally obtain a higher payoff. Salamanca (2016) studies the informed party that is able to choose and commit to the mediation

plan. The author characterizes sender’s value function as the concavification of sender’s indirect virtual utility function over prior beliefs. This paper can be seen as the case of complete alignment between sender’s and mediator’s preferences. In my paper, if the information source and the mediator are aligned, then the only equilibrium is completely uninformative. [Skreta and Perez-Richet \(2021\)](#) consider a game with a similar extensive form to this paper. The mediator designs a test to detect the costly falsification of the state of the world by the sender.⁵ The main difference of [Skreta and Perez-Richet \(2021\)](#) and this paper is the payoff structure: the mediator designs the test to maximize the receiver’s payoff while the sender wishes the receiver to take a certain action in [Skreta and Perez-Richet \(2021\)](#).

This paper also contributes to the *constrained information design* literature. This literature seeks to extend the standard Bayesian persuasion framework of [Kamenica and Gentzkow \(2011\)](#) by adding meaningful constraints the persuading side has to face. [Le Treust and Tomala \(2019\)](#) and [Tsakas and Tsakas \(2019\)](#) study the setup where the persuading side communicates with the receiver through a channel that is subject to exogenous noise. The optimal payoff is characterized as a function of the Shannon channel capacity in [Le Treust and Tomala \(2019\)](#).⁶ [Lipnowski and Mathevet \(2018\)](#) impose the behavioral assumptions on the receiver that leads to non-Bayesian updating and analyze the optimal information disclosure. Compared to these papers, the media in the role of a persuading side face novel constraints capturing the media’s inability to access the state directly. Instead, the media incentivize the source to supply information by carefully designing the information protocol.

Finally, [Boleslavsky and Kim \(2018\)](#) consider the setup where the agent exerts a privately observed effort that determines the state distribution. Thus, the persuader has to not only persuade the receiver to take some action, but also incentivize the agent’s effort. This paper has the additional constraints in the form of moral hazard, whereas in my paper the honesty

⁵[Skreta and Perez-Richet \(2021\)](#) cover the case of the costless falsification that corresponds to cheap-talk messages.

⁶The technique developed in [Le Treust and Tomala \(2019\)](#) and [Doval and Skreta \(2018\)](#) corresponds to rewriting the additional constraints as a function of receiver’s posterior beliefs distribution. However, in my problem, the state space is continuous and honesty constraints have to be satisfied for every pair of states. Even though the inequalities corresponding to honesty conditions can be written in accordance with [Doval and Skreta \(2018\)](#), their method does not make the problem tractable.

constraints correspond to the adverse selection problem. Moreover, the agent's problem can be summarized by a single equation that can be written as the expectation over the distribution of posterior beliefs, so that the methodology of [Doval and Skreta \(2018\)](#) is applicable. [Ball \(2020\)](#) introduces the model of predictive scoring. The scoring agency with commitment power aggregates multiple features of the sender into a score. The sender's features are correlated with the state that the receiver wishes to match. The sender is able to distort her features at a cost. The scoring agency is aligned in preferences with the receiver. The optimal scoring rule puts smaller weights on some features to defer an excessive distortion by the sender.

3 Model

This section introduces a game between an informed elite, which I call a *source* (S , she), a state-owned *media* (M , it), and an uninformed *receiver* (R , he).

The receiver has to decide whether to undertake the status-quo action a_0 , or the mobilizing action a_1 . The mobilizing action corresponds to some political objective of the ruler. The payoff of the receiver $u_R(a, \theta)$ depends on his action $a \in A = \{a_0, a_1\}$ and the ruler's competence $\theta \in \Theta = [0, 1]$. The ruler's competence is unknown to the receiver but he holds a prior μ_0 on Θ that is common to all the players. The receiver's preference parameter $\delta_R(\theta) = u_R(a_1, \theta) - u_R(a_0, \theta)$ captures the receiver's net payoff from the mobilizing action, and the function $\delta_R(\theta)$ is assumed to be strictly increasing. That is, the receiver prefers the mobilizing action if the ruler's competence is high enough. Moreover, I assume the following *tension* condition:

$$\int_0^1 \delta_R(\theta) d\mu_0 < 0. \quad (1)$$

This condition indicates that under the prior the receiver opts for the status-quo action.

A source perfectly learns the ruler's competence and cares about the receiver's action. The source is referred to as type- θ source if she learns that the ruler's competence is θ . The source's payoff function $u_S(a, \theta)$ is summarized by two measurable sets representing source's ordinal preferences, $\Theta_0 = \{\theta \in [0, 1] : u_S(a_0, \theta) > u_S(a_1, \theta)\}$ and $\Theta_1 = \{\theta \in [0, 1] : u_S(a_1, \theta) > u_S(a_0, \theta)\}$. In words, Θ_0 captures the source types that strictly prefer the status-quo action, whereas Θ_1 captures the source types that strictly prefer the mobilizing action. The measure

of types that are indifferent between a_0 and a_1 is assumed to be zero.⁷ The source can only communicate with the receiver indirectly, by sending a costless message $m \in \mathcal{M}$ to the state-owned media. The set of messages $\mathcal{M}$ has at least as many elements as Θ .

The state-owned media wish to promote the ruler's interests. In particular, the media want the receiver to undertake the mobilizing action irrespective of the ruler's competence. The media's payoffs of the status-quo action and the mobilizing action are normalized to 0 and 1, respectively. Therefore, the media's expected payoff is simply the probability of the mobilizing action being chosen. The media can communicate with the receiver but cannot generate information itself. Instead, information has to be provided to the media by the source in the form of message $m \in \mathcal{M}$. The media then produce a costless report $r \in \mathcal{R}$ observed by the receiver. The set of reports $\mathcal{R}$ has at least two elements.

I assume that the media can commit to how the reports are generated based on the messages provided by the source. In particular, at the beginning of the game, the state-owned media publicly choose an editorial policy $\pi : \mathcal{M} \rightarrow \Delta(\mathcal{R})$, where $\pi(r|m)$ is the probability of generating report r after observing the source's message m . I refer to the editorial policy π as a strategic dissemination protocol, or simply a *protocol*.

Timing.—I summarize the timing of the game. The game starts with the state-owned media committing to the strategic dissemination protocol, $\pi : \mathcal{M} \rightarrow \Delta(\mathcal{R})$, observed by all players. The ruler's competence $\theta \in [0, 1]$ then realizes as the draw from the distribution μ_0 . The source observes θ and π and sends a costless message $m \in \mathcal{M}$ to the media. The report $r \in \mathcal{R}$ is then generated by the media as the draw from the distribution $\pi(\cdot|m)$. Finally, the receiver observes the protocol π and the report r , forms the posterior belief μ , and decides whether to undertake the status-quo action a_0 or the mobilizing action a_1 . The payoffs then are realized.

Equilibrium.—An *equilibrium* consists of four measurable maps: a messaging strategy $\rho : \Theta \rightarrow \Delta(\mathcal{M})$ for S , an information dissemination protocol $\pi : \mathcal{M} \rightarrow \Delta(\mathcal{R})$ for M , a probability of choosing the mobilizing action $\alpha : \mathcal{R} \rightarrow [0, 1]$ for R , and a belief mapping $\mu : \mathcal{R} \rightarrow \Delta(\Theta)$ for R . An equilibrium is the protocol π chosen by M and a perfect Bayesian equilib-

⁷If the measure of indifferent source types is nonzero, the media's problem is relaxed in the sense that the media have to satisfy fewer incentive constraints on the source's side.

rium of the subgame that follows the M 's choice. Specifically, given π , a perfect Bayesian equilibrium (PBE) is a tuple (ρ, α, μ) that satisfies

1. (belief formation)

μ is obtained from μ_0 via Bayes' rule, given ρ , whenever well-defined;

2. (receiver's best-response)

$\alpha(r) = 1$ if $\int_{\Theta} \delta_R(\cdot) d\mu(\cdot|r) > 0$, and $\alpha(r) = 0$ if $\int_{\Theta} \delta_R(\cdot) d\mu(\cdot|r) < 0$;

3. (sender's best-response)

$\rho(\theta)$ is supported on $\operatorname{argmax}_{m \in \mathcal{M}} \int_{\mathcal{R}} [u_S(a_1, \theta) \alpha(\cdot) + u_S(a_0, \theta)(1 - \alpha(\cdot))] d\pi(\cdot|m)$ for every $\theta \in \Theta$.

Following the information design literature⁸, the agent with a commitment power is assumed to be able to steer other agents toward her favorite PBE. Thus, for every π , M chooses a PBE that generates the highest media's ex ante payoff denoted as $V(\pi)$. Finally, the equilibrium π is chosen to maximize $V(\pi)$. I denote the value function of the media as V . This completes the definition of the equilibrium.

4 Equilibrium Analysis

I analyze the model using revelation principle. It is without loss to focus on the direct protocols where the source truthfully reports the ruler's competence and the receiver obediently follows an action recommendation. I characterize the set of the direct protocols that satisfy honesty and obedience conditions. Finally, I solve for the optimal protocol for the media and discuss its properties.

A protocol π is said to be direct if $\mathcal{M} = \Theta$ and $\mathcal{R} = A$. That is, for a direct protocol π , the source is asked to report a competence level θ and the media make a binary action recommendation to the receiver.

⁸See Kamenica and Gentzkow (2011), Alonso and Câmara (2016), and Bergemann and Morris (2016) among others. Mathevet et al. (2020) develop the methodology of analyzing persuasion problems for various equilibrium selection rules.

In a direct protocol, S is said to be *honest* if it is optimal for her to report the ruler's competence truthfully. R is said to be *obedient* if it is optimal for him to follow a recommendation. A direct protocol $\pi : \Theta \rightarrow \Delta(A)$ is Bayesian incentive-compatible if S is honest and R is obedient. Specifically, S is honest given R 's obedience if

$$u_S(a_1, \theta)\pi(a_1|\theta) + u_S(a_0, \theta)\pi(a_0|\theta) \geq u_S(a_1, \theta)\pi(a_1|\theta') + u_S(a_0, \theta)\pi(a_0|\theta') \quad (2)$$

for every $\theta, \theta' \in \Theta$. I call a direct protocol π *honest* if it satisfies (2).

R is obedient given S 's honesty if following recommendation a_1 is optimal, that is,

$$\int_0^1 \delta_R(\theta)\pi(a_1|\theta)d\mu_0 \geq 0, \quad (3)$$

and following recommendation a_0 is optimal, that is,

$$-\int_0^1 \delta_R(\theta)\pi(a_0|\theta)d\mu_0 \geq 0. \quad (4)$$

Note that the tension condition (1) and the inequality (3) imply the inequality (4). Then I call a direct protocol π *obedient* if it satisfies (3).

The revelation principle states that without loss the media can focus on the direct protocols that are honest and obedient.

Lemma 1. *Given any PBE in the original game followed by π , there exists an incentive-compatible direct protocol π^* under which the media get the same expected utility when S is honest and R is obedient as in this PBE.*

All proofs are in the Appendix. By Lemma 1 an optimal incentive-compatible direct protocol is also an equilibrium in the class of all possible protocols. From now on I focus on the characterization of the optimal incentive compatible direct protocol.

It turns out that the honesty constraint significantly simplifies the problem by disciplining any incentive-compatible direct protocol. Lemma 2 summarizes this observation.

Lemma 2. *A direct protocol π is honest if and only if there exist π_0 and π_1 , with $\pi_0 \leq \pi_1$, such that $\pi(a_1|\theta) = \pi_0$ for every $\theta \in \Theta_0$, and $\pi(a_1|\theta) = \pi_1$ for every $\theta \in \Theta_1$.*

Thus, an honest direct protocol is characterized by a pair of numbers $\pi_i, i \in \{0, 1\}$, capturing the probability that the receiver takes the mobilizing action when the ruler's competence is in Θ_i . Intuitively, the source will always attempt to induce the highest probability of her preferred action. Therefore, the probability of a_i prescribed by protocol π has to be identical across all $\theta \in \Theta_i$. Furthermore, the source types that prefer action a_1 have to be provided with a higher probability π_1 of implementing this action compared to probability π_0 provided to the types that prefer action a_0 .

To find the set of incentive-compatible protocols, I combine the insight of Lemma 2 with the obedience condition. To this end, define $I_0 = \int_{\Theta_0} \delta_R(\theta) d\mu_0$ and $I_1 = \int_{\Theta_1} \delta_R(\theta) d\mu_0$. In words, I_0 and I_1 capture receiver's preferences in conjunction with source's preferences. It turns out that these statistics are sufficient to pin down the set of incentive-compatible protocols. Note that $\frac{I_i}{\mu_0(\Theta_i)}, i \in \{0, 1\}$, is the expectation of the receiver's net payoff from the mobilizing action conditional on the competence being in Θ_i , $\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_i]$. Proposition 1 characterizes the set of incentive-compatible direct protocols depending on the signs of the conditional expectations $\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_0]$ and $\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_1]$.

Proposition 1. *The set of incentive-compatible direct protocols $\mathcal{I}$ is characterized by (π_0, π_1) , $\pi_0 \leq \pi_1$, where π_i is the probability of implementing action a_1 following any $\theta \in \Theta_i$. Furthermore,*

- *if $I_0 < 0$ and $I_1 \geq 0$, then $\mathcal{I} = \{(\pi_0, \pi_1) \in [0, 1]^2 : \pi_1 I_1 + \pi_0 I_0 \geq 0\}$;*
- *otherwise, $\mathcal{I} = \{(0, 0)\}$.*

To get the intuition behind this result, first, consider the case of $\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_1] < 0$. In words, this inequality means that there is a sufficiently large portion of the source types who disagree with the receiver on the preferable action: they prefer a_1 , whereas the receiver would choose a_0 if the ruler's competence was known. The receiver is too skeptical of the statements of the source closely aligned with the media in trying to assure the mobilizing action.⁹ Thus, in

⁹By the standard intuition from sender-receiver games, if the conflict between sender's and receiver's preferences is too large, then no meaningful information can be transmitted. For example, in the uniform-quadratic setup of the cheap-talk game of Crawford and Sobel (1982), if the sender's bias is too large, then the equilibrium is necessarily completely uninformative.

this case the only incentive-compatible protocol involves pooling: $\pi_0 = \pi_1 = 0$.¹⁰ Otherwise, the information about whether the ruler's competence is in Θ_0 or Θ_1 can be meaningfully transmitted to the receiver. Furthermore, Proposition 1 shows that in this case the obedience constraint implies the requirement $\pi_0 \leq \pi_1$ of the honesty constraint.

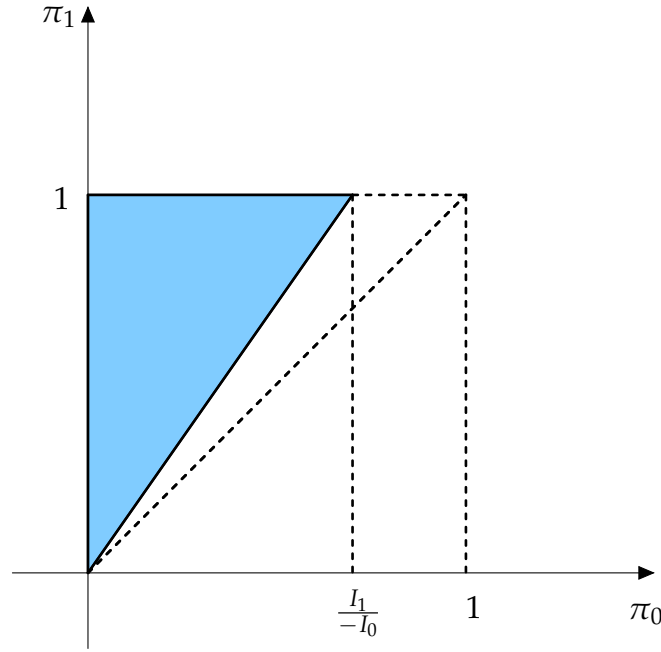


Figure 1: The set of incentive-compatible direct protocols, when $\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_0] < 0$ and $\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_1] > 0$.

Figure 1 depicts the set of incentive-compatible protocols for the case when complete pooling is not the only incentive-compatible protocol. This figure is similar to the set of obedient protocols in the standard Bayesian persuasion problem with a binary state (see, for example,

¹⁰Note that even though $\pi_0 = \pi_1 = 0$, communication is not necessarily uninformative. However, all receiver's posterior beliefs generated by this protocol are low enough so that the status-quo action is always taken.

Bergemann and Morris (2019)). Here, however, the state space is continuous. The honesty constraints discipline the protocol over the states in Θ_0 and Θ_1 . Thus, the binary state in the standard Bayesian persuasion problem can be seen as whether the state in my problem lies in Θ_0 or Θ_1 .

Proposition 1 paves the way to finding the optimal media's protocol. Indeed, given the previous insights, the media's problem can be written as follows:

$$V = \max_{(\pi_0, \pi_1) \in \mathcal{I}} \{ \mu_0(\Theta_0) \cdot \pi_0 + \mu_0(\Theta_1) \cdot \pi_1 \}.$$

Thus, the media maximize the linear function over the set of incentive-compatible protocols $\mathcal{I}$ defined by the linear inequalities and given by Proposition 1. Then there necessarily is an extreme point solution. Proposition 2 finds this extreme point.

Proposition 2. *If $I_0 < 0 \leq I_1$, then the solution to the media problem is the pair (π_0, π_1) such that $\pi_0 = \pi(a_1|\theta)$ for every $\theta \in \Theta_0$, $\pi_1 = \pi(a_1|\theta)$ for every $\theta \in \Theta_1$, and $(\pi_0, \pi_1) = \left(\frac{I_1}{-I_0}, 1\right)$. The ex ante media's payoff is*

$$V = \mu_0(\Theta_1) \cdot \frac{\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_1] - \mathbb{E}[\delta_R(\theta)|\theta \in \Theta_0]}{-\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_0]}.$$

If $I_0 < 0 \leq I_1$ is not satisfied then $\pi_0 = \pi_1 = 0$ and the media's payoff is 0.

When the conflict between source's and the receiver's preferences is too large, the media cannot do better than making the protocol completely uninformative. Otherwise, the media signal whether the ruler's competence lies in Θ_0 or Θ_1 . From the receiver's perspective, if the receiver gets the recommendation of the status-quo action, then he knows with certainty that the ruler's competence is in Θ_0 . On the other hand, the recommendation of the mobilizing action can come from any competence level, but the protocol renders the probabilities in the exact way to make the receiver indifferent between two actions.

4.1 Example

In order to gain insight about properties of the media-optimal protocol, I impose the functional forms for the players' payoffs. I assume that the receiver's payoff function is $u_R(\theta, a) = 1\{a =$

$a_1\} \cdot (\theta - \omega)$, where $\omega \in [0, 1]$ is interpreted as receiver's pickiness commonly known to the players. The pickiness level corresponds to how critical the receiver is of the government. A pickier receiver would require a higher ruler's competence to oblige with choosing the mobilizing action. If the ruler's competence surpasses ω , then the receiver prefers the mobilizing action. The receiver's net payoff from the mobilizing action is then simply a linear function $\delta_R(\theta) = \theta - \omega$. By the tension condition (1), $\omega > \frac{1}{2}$. If $\omega \leq \frac{1}{2}$, then the media make the protocol completely uninformative and extracts the payoff of 1.

Furthermore, I assume that Θ_0 and Θ_1 are half-intervals: $\Theta_0 = [0, \bar{\theta})$ and $\Theta_1 = (\bar{\theta}, 1]$, where $\bar{\theta} \in [0, 1]$ is the relevant source's payoff parameter commonly known to the players. I refer to $\bar{\theta}$ as a source's *threshold*. That is, all the source types above $\bar{\theta}$ strictly prefer the mobilizing action, whereas all the source types below $\bar{\theta}$ strictly prefer the status-quo action. The source with type $\bar{\theta}$ is indifferent between a_0 and a_1 . If $\bar{\theta} = 0$, then the source is aligned in preferences with the media. If $\bar{\theta} = \omega$, then the source is aligned in preferences with the receiver. Finally, for the sake of exposition, let the prior μ_0 be the uniform distribution on $[0, 1]$.

Note that in the general case covered before, there may coexist two sets of the source types that do not want their information to be revealed: types in Θ_0 that know that the ruler's competence exceeds ω and types in Θ_1 that know that the ruler's competence is below ω . The example leaves only one of these sets present depending on the relation between $\bar{\theta}$ and ω . Threshold $\bar{\theta}$ can be interpreted as the source's pickiness level that is potentially different from the receiver's ω .

For this example, the summary statistics I_0 and I_1 of source's and the receiver's preferences can be directly calculated as

$$I_0 = \int_0^{\bar{\theta}} (\theta - \omega) d\theta = \frac{\bar{\theta}(\bar{\theta} - 2\omega)}{2},$$

$$I_1 = \int_{\bar{\theta}}^1 (\theta - \omega) d\theta = \frac{(1 - \bar{\theta})(1 + \bar{\theta} - 2\omega)}{2}.$$

Then Proposition 2 readily establishes the optimal protocol and the media's ex ante payoff from this protocol.

Claim 1. *If $\bar{\theta} < 2\omega - 1$, then the optimal protocol is $\pi_0 = \pi_1 = 0$, resulting in the media's*

payoff of 0. If $\bar{\theta} \geq 2\omega - 1$, then the optimal protocol is

$$(\pi_0, \pi_1) = \left(\frac{(1 - \bar{\theta})(1 + \bar{\theta} - 2\omega)}{\bar{\theta}(2\omega - \bar{\theta})}, 1 \right),$$

and the media's payoff is

$$V = \frac{1 - \bar{\theta}}{2\omega - \bar{\theta}}.$$

Figure 2 shows the media's optimal protocol and the value function for $\omega = \frac{2}{3}$ for various source's thresholds $\bar{\theta}$.

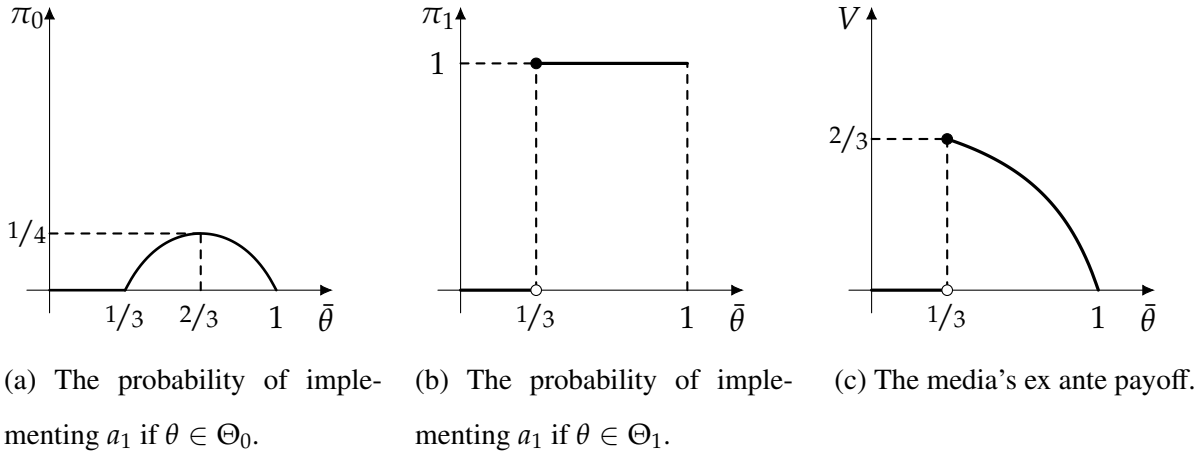


Figure 2: The solution to the media's problem for $\omega = 2/3$ as a function of $\bar{\theta}$.

Given Claim 1, it is straightforward to derive the relevant comparative statics with respect to $\bar{\theta}$ and ω . A receiver with pickiness ω' is said to be more aligned with the media than a receiver with pickiness ω'' if $\omega' < \omega''$. Similarly, a source with threshold $\bar{\theta}'$ is said to be more aligned with the media than a source with threshold $\bar{\theta}''$ if $\bar{\theta}' < \bar{\theta}''$.

Claim 2. *The media get the ex ante payoff of 0 if $\bar{\theta} < 2\omega - 1$. As long as $\bar{\theta} \geq 2\omega - 1$, the media's ex ante payoff increases if the receiver or the source become more aligned with the media.*

It is easier for the media to persuade a more aligned receiver to undertake the mobilizing action. For a more aligned source, there are two effects. First, the measure of types Θ_1 that guarantees the implementation of action a_1 goes up. Second, for the source types from Θ_0 , the probability of implementing the mobilizing action is increasing in $\bar{\theta}$ for $\bar{\theta} < \omega$ and decreasing

for $\bar{\theta} > \omega$. When π_0 is increasing in $\bar{\theta}$, the first effect outweighs the second effect. Therefore, the media's payoff is lower for the pickier source as long as $\bar{\theta} \geq 2\omega - 1$. To summarize, if the source is closely aligned with the media, $\bar{\theta} < 2\omega - 1$, the media's payoff stays at zero. When $\bar{\theta}$ reaches the level of $2\omega - 1$, the media's payoff jumps to $2 - 2\omega$ and then starts to decrease to 0 with increasing $\bar{\theta}$. This effect is illustrated in Figure 2(c).

Observe that the receiver's ex ante payoff is always zero irrespective of the source's preferences. Indeed, when the protocol never recommends the mobilizing action, the receiver undertakes the status-quo action and gets zero payoff. When the source is not too closely aligned with the media, the media ensure that the receiver is kept indifferent between the status-quo and mobilizing actions upon receiving the recommendation of the mobilizing action, so that the ex ante payoff of the receiver is again zero.

Claim 3 establishes a comparative statics of the source's payoff as a function of the receiver's pickiness.

Claim 3. *Suppose that $\bar{\theta} \geq 2\omega - 1$. The source types from Θ_1 get their favorite action a_1 with probability 1 irrespective of ω . The source types from Θ_0 get their favorite action a_0 with probability $1 - \pi_0$ that is increasing in ω . That is, the source types from Θ_0 are worse off when the receiver is more aligned with the media.*

Hence, by Claim 3, ex ante (before learning the ruler's competence) the source is worse off when the receiver is more aligned with the media. Indeed, the source types from Θ_1 always get their favorite action. The source types from Θ_0 benefit from a receiver with greater ω as it becomes harder for the media to persuade the receiver to undertake the mobilizing action.

Finally, I compare the media's optimal protocol to the protocol in the standard Bayesian persuasion problem, that is, the problem with no honesty constraints.

Claim 4. *The solution to the media's problem facing no honesty constraints is*

$$\pi(a_1|\theta) = \begin{cases} 1, & \text{if } \theta \geq 2\omega - 1, \\ 0, & \text{otherwise.} \end{cases}$$

The media's ex ante payoff is equal to $2 - 2\omega$.

The optimal protocol with honesty concerns achieves the payoff of the media facing no honesty constraints when $\bar{\theta} = 2\omega - 1$. In other words, the media-optimal source's threshold corresponds to the threshold on the Bayesian persuasion protocol that renders the receiver to be indifferent between a_0 and a_1 . If the source's threshold $\bar{\theta}$ falls below $2\omega - 1$, the media's payoff drops to zero. This discontinuity is the consequence of the discontinuity of the set of incentive-compatible protocols in $\bar{\theta}$. When $\bar{\theta} < 2\omega - 1$, only the protocol that never recommends a_1 is available. At $\bar{\theta} = 2\omega - 1$, the Bayesian persuasion protocol becomes available and is employed by the media. Importantly, even when $\bar{\theta} = 2\omega - 1$, the media facing no honesty constraints have access to a larger set of incentive-compatible protocols. However, in this case the solution to my problem and Bayesian persuasion problem coincide.

5 Persuading the Public

This section allows the receiver to have private information. The media attempt to persuade the population of receivers to choose the mobilizing action. The media's report r is publicly revealed to the unit mass of receivers. Each receiver cares only about his own action $a_i \in \{a_0, a_1\}$.¹¹ I impose the same assumptions on the payoff functions as in Section 4.1. That is, $\Theta_0 = [0, \bar{\theta})$ and $\Theta_1 = (\bar{\theta}, 1]$. The ruler's competence level θ is assumed to be a draw from the uniform distribution on $[0, 1]$. The receiver i 's net payoff from the mobilizing action is a linear function $\delta_R(\theta, \omega_i) = \theta - \omega_i$, where the receiver's pickiness $\omega_i \in [0, 1]$ is the receiver i 's private information. The mass of the receivers with the pickiness below or equal to ω is captured by an absolutely continuous cumulative distribution function H , with a strictly positive on $(0, 1)$ density h . Denote the measure of a set $\Omega \subseteq [0, 1]$ generated by H as $\eta(\Omega)$. Type- θ source extracts $u_S(a_1, \theta)$ from each receiver taking the mobilizing action and $u_S(a_0, \theta)$ from each receiver taking the status-quo action. The timing is unchanged. However, the source and the media have to evaluate their payoffs as the expectation over the receiver's types. The media's goal is to maximize the proportion of the receivers that choose the mobilizing action.

This setup is isomorphic to the problem with a single receiver having private information

¹¹In a voting application, this assumption corresponds to the sincere voting paradigm. For example, see [Alonso and Câmara \(2016\)](#).

about ω . The common prior on ω is captured by the distribution H , and θ and ω are assumed to be independent. I assume that the media cannot elicit private information from the receiver. Then the analogue of the revelation principle for this case can be shown. Instead of an unconditional action recommendation, the media now offer a contingent recommendation, i.e., an action recommendation for each receiver's type. A typical contingent recommendation has the form of $\omega \mapsto \{a_0, a_1\}$. Thus, without loss, it can be seen as the subset Ω_1 of receivers that are recommended to choose a_1 . To this end, I focus on a protocol $\phi : \Theta \rightarrow \Delta(\mathcal{P}([0, 1]))$, where $\mathcal{P}([0, 1])$ is the set of measurable subsets of the unit interval. I will show that only recommendations of the form $[0, b]$, $b \in [0, 1]$, are sent. Let $\text{supp}\phi$ denote the set of contingent recommendations that appear in the protocol ϕ after some reported competence level θ , $\phi(\cdot|\theta) > 0$, and assume that $\text{supp}\phi$ is finite.¹²

A contingent protocol $\phi : \Theta \rightarrow \Delta(\mathcal{P}([0, 1]))$ is Bayesian incentive-compatible if S 's honesty and R 's obedience form an equilibrium. Specifically, S is honest given R 's obedience if

$$\sum_{\Omega_1 \in \text{supp}\phi} [\eta(\Omega_1)u_S(a_1, \theta) + (1 - \eta(\Omega_1))u_S(a_0, \theta)] (\phi(\Omega_1|\theta) - \phi(\Omega_1|\theta')) \geq 0 \quad (5)$$

for every $\theta, \theta' \in \Theta$, that is, it is optimal for the source to report the ruler's competence. I call ϕ honest if it satisfies (5).

R is obedient given S 's honesty if, for every $\Omega_1 \in \text{supp}\phi$,
for every $\omega \in \Omega_1$,

$$\int_0^1 \delta_R(\cdot, \omega) d\phi(\Omega_1|\cdot) \geq 0, \quad (6)$$

for every $\omega \in [0, 1] \setminus \Omega_1$,

$$\int_0^1 \delta_R(\cdot, \omega) d\phi(\Omega_1|\cdot) \leq 0. \quad (7)$$

I call ϕ obedient if it satisfies (6) and (7).

The following lemma shows that an optimal incentive-compatible contingent protocol is also an equilibrium in the class of all possible protocols.

¹²In principle, a protocol ϕ can have an infinite support. Then the summation in the honesty constraint (5) has to be substituted by appropriate integration.

Lemma 3. *Given any PBE in the original game followed by π , there exists an incentive-compatible contingent protocol $\phi : \Theta \rightarrow \Delta(\mathcal{P}([0,1]))$ under which the media get the same expected utility when S is honest and R is obedient as in this PBE.*

As in the setup with a known receiver's type, the incentive constraints reduce the dimension of the search for the optimal protocol. I start by simplifying the obedience constraint. Lemma 4 shows that any contingent recommendation is necessarily an interval. The obedience constraint can be reduced to a single equation for each recommendation in the support. This equation requires that the receiver with the highest pickiness level among the receivers that are recommended a_1 has to be indifferent between the mobilizing and status-quo actions.

Lemma 4. *Every obedient protocol $\phi : \Theta \rightarrow \Delta(\mathcal{P}([0,1]))$ exclusively sends contingent recommendations of the following form: a_1 is recommended to receivers with $\omega \in [0, b]$, $b \in [0, 1]$; otherwise, a_0 is recommended. For every $b \in \text{supp}\phi$, the obedience constraint is summarized by*

$$\int_0^1 (\theta - b) d\phi([0, b]|\theta) = 0.$$

Lemma 4 illustrates that the contingent recommendations are intuitive: only receivers that are sufficiently aligned with the media are recommended to take the mobilizing action. Lemma 5 provides the further simplification that mirrors Lemma 2 and characterizes honest protocols.

Lemma 5. *A protocol $\phi : \Theta \rightarrow \Delta(\mathcal{P}([0,1]))$ is honest if and only if there exist s_0 and s_1 , with $s_0 \leq s_1$, such that for every $\theta \in \Theta_i$, $i \in \{0, 1\}$,*

$$s_i = \sum_{[0, b] \in \text{supp}\phi} \phi([0, b]|\theta) H(b).$$

As before, the source types within sets Θ_0 and Θ_1 have to be provided the same probability of their preferred action being implemented. This probability has to be evaluated as an expectation over the receiver's private information. The source types from Θ_1 get the favorite mobilizing action with probability s_1 , and the source types from Θ_0 get the undesirable mobilizing action with probability s_0 . Naturally, for the honesty constraint to hold, the media ensure that $s_1 \geq s_0$.

Combining the results of Lemmas 3, 4, and 5, the media's problem can be written as the following Lemma 6 prescribes.

Lemma 6. *The media's problem is*

$$V = \max_{\phi, s_0, s_1} \{ \bar{\theta} \cdot s_0 + (1 - \bar{\theta}) \cdot s_1 \},$$

subject to

$$s_0 = \sum_{[0, b] \in \text{supp}\phi} \phi([0, b]|\theta) H(b)$$

for every $\theta \in [0, \bar{\theta})$,

$$s_1 = \sum_{[0, b] \in \text{supp}\phi} \phi([0, b]|\theta) H(b)$$

for every $\theta \in [\bar{\theta}, 1]$; $s_0 \leq s_1$; and

$$\int_0^1 (\theta - b) d\phi([0, b]|\theta) = 0$$

for every $[0, b] \in \text{supp}\phi$.

This problem is still complicated, since the optimal protocol can potentially provide multiple contingent recommendations given each reported competence level θ . I do not describe the solution to the problem stated in Lemma 6 explicitly. Instead, I deliver the lower bound on the media's payoff by finding the optimal protocol within the restricted class of protocols. Define a *simple protocol* as a protocol with a support having at most two elements, $[0, \bar{b}]$ and $[0, \underline{b}]$, with $\bar{b} \geq \underline{b}$. In what follows, I characterize the optimal simple protocol. I will show the condition on the distribution of receiver's types H , under which the media's payoff achieves this lower bound.

For a simple protocol, the honesty constraints in Lemma 6 can be written as follows:

$$s_0 = H(\underline{b}) + \phi_0(H(\bar{b}) - H(\underline{b})),$$

where $\phi_0 = \phi([0, \bar{b}]|\theta)$ for every $\theta \in \Theta_0$, and

$$s_1 = H(\underline{b}) + \phi_1(H(\bar{b}) - H(\underline{b})),$$

where $\phi_1 = \phi([0, \bar{b}]|\theta)$ for every $\theta \in \Theta_1$. It has to be the case that $\phi_0 \leq \phi_1$. For a simple protocol, the honesty constraints specify that the probability of generating the “larger” contingent recommendation $[0, \bar{b}]$ has to be constant within the competence levels in Θ_0 and Θ_1 and it has to be higher for the sources in Θ_1 .

The obedience constraint then pins down $\bar{b}$ and $\underline{b}$ as a function of ϕ_0 and ϕ_1 . Lemma 7 establishes the bounds on $\bar{b}$ and $\underline{b}$ and shows that, for any pair $(\underline{b}, \bar{b})$ within these bounds, there exists a simple protocol with the support on $[0, \underline{b}]$ and $[0, \bar{b}]$. The operator $\mathbb{E}$ is the expectation with respect to the prior distribution on Θ .

Lemma 7. *Every simple incentive-compatible protocol $\phi : [0, 1] \rightarrow \Delta(\{[0, \underline{b}], [0, \bar{b}]\})$ has to satisfy $\bar{b} \in [\mathbb{E}[\theta], \mathbb{E}[\theta|\theta \in \Theta_1]]$ and $\underline{b} \in [\mathbb{E}[\theta|\theta \in \Theta_0], \mathbb{E}[\theta]]$. For every pair $(\underline{b}, \bar{b})$ within these bounds, there exists an incentive-compatible simple protocol with the support on $[0, \underline{b}]$ and $[0, \bar{b}]$, such that $\phi_0 = \phi([0, \bar{b}]|\theta)$ for every $\theta \in \Theta_0$, $\phi_1 = \phi([0, \bar{b}]|\theta)$ for every $\theta \in \Theta_1$, with $\phi_0 \leq \phi_1$.*

From Lemma 7, every simple protocol is characterized by the pair of numbers $\bar{b}$ and $\underline{b}$. This observation paves the way to finding the optimal simple protocol. Proposition 3 provides the geometric characterization of the solution to the media's problem. Let $\text{cav}H$ be the concavification of H , that is, the smallest concave function that majorizes H . Let $\hat{H}$ be the function H reduced to the domain $[\mathbb{E}[\theta|\theta \in \Theta_0], \mathbb{E}[\theta|\theta \in \Theta_1]]$. Proposition 3 shows that the optimal simple protocol delivers the media the payoff equal to the concavification of $\hat{H}$ evaluated at $\mathbb{E}[\theta]$.

Proposition 3. *The media's payoff from the optimal simple protocol is equal to $\text{cav}\hat{H}[\mathbb{E}[\theta]]$.*

Proposition 3 obtains the lower bound on the media's equilibrium payoff. The upper bound on the media's payoff is given by $\text{cav}H[\mathbb{E}[\theta]]$. Indeed, the receiver ultimately bases his decision on the posterior mean of the ruler's competence. If it was possible for the media to induce every distribution of posterior means whose expectation is the prior mean, then the solution to the media's problem would correspond to $\text{cav}H[\mathbb{E}[\theta]]$. However, this is not always feasible (see, for example, Gentzkow and Kamenica (2016)). If those bounds are equal to each other, then the simple protocol is optimal across all incentive-compatible protocols. Corollary 1 summarizes this observation.

Corollary 1. *The simple protocol is optimal for the media if $\text{cav}\hat{H}[\mathbb{E}[\theta]] = \text{cav}H[\mathbb{E}[\theta]]$.*

Note that this sufficient condition can be checked just by knowing primitives of the model: distribution H , prior distribution on Θ , and source's preferences. The immediate consequence

of Corollary 1 is that for concave H , the simple protocol is optimal. Moreover, $\bar{b} = \underline{b}$, that is, the optimal protocol is uninformative.

Finally, to illustrate the bound provided in Proposition 3, I consider a unimodal distribution of receiver's types that recently gained a lot of attention in the literature, namely, Kolotilin et al. (2017), Lipnowski et al. (2019), and Shishkin (2019).

The unimodal distribution corresponds to the density strictly increasing until reaching mode m and then strictly decreasing. As a result, the corresponding cumulative distribution function is convex-concave. The black line in Figure 3 illustrates the example of such distribution.

With no honesty concerns, Kolotilin et al. (2017) show that the optimal policy is upper censorship, when the distribution of receiver's types is unimodal: it reveals all states below and pools all states above a threshold.¹³ The upper-censorship policy in my setup corresponds to the protocol ϕ , with $\phi([0, \theta]|\theta) = 1$ for $\theta < t$ and $\phi([0, b]|\theta) = 1$ for $\theta \geq t$, where $t > 0$ is a threshold and $b = \mathbb{E}[\theta|\theta \geq t]$. However, the upper-censorship policy is not incentive-compatible, when the honesty constraints are present. Indeed, pick two types $\theta', \theta'' \in \Theta_0 \cap [0, t)$, $\theta' < \theta''$ and consider the incentives of type- θ'' source under the upper-censorship. Reporting θ' gives the source probability $1 - H(\theta')$ of the status-quo action chosen, whereas reporting θ'' produces probability $1 - H(\theta'')$. Type- θ'' source prefers to misreport, and the honesty constraint is not satisfied. Indeed, by Lemma 5 the sources preferring the status-quo action have to be provided with the same probability of this action implemented. Under the upper-censorship policy, each source type from Θ_0 wants to mimic the lowest possible type, so that the status-quo action is always chosen. This observation leads me to expect some pooling for the low states in the optimal protocol.

¹³The mode of the distribution has to be sufficiently large. Otherwise, the uninformative policy is optimal as most of the receivers are closely aligned with the media.

6 Concluding Remarks

This paper presents a model of information disclosure by a state-owned media to an uninformed receiver choosing between two actions. The problem is that the media do not have direct access to relevant information. Instead, it has to be supplied by the informed elite having interests in the receiver's action. Therefore, the optimal media's editorial policy has to not only convince the receiver to undertake the media-favorite action, but also cater to the elite's preferences to incentivize the information supply. I show how these additional incentive constraints shape the optimal editorial policy and outline the welfare implications of this policy. I show the conditions under which the honesty constraints are binding and there is a meaningful communication depending on the preference parameters. I close with the discussion of assumptions that are substantial for my results.

Discussion of assumptions.— I assume that there are no transfers between players. In this sense, I study the purely informational model of the interaction between the source, the media, and the receiver. In reality, the source may be paid for promoting the ruler's competence or the receiver may be paid by the ruler for undertaking the mobilizing action. I leave this possibility out of the model.

I assume that the media have commitment power. As explained in [Gehlbach and Sonin \(2014\)](#) and [Gentzkow et al. \(2015\)](#), the editorial policy cannot be easily changed and consistent bias in reporting is detected by receivers. This commitment assumption may be relaxed in the fashion of [Lipnowski et al. \(2019\)](#), where with some probability the media may secretly change the editorial policy after observing the source's message. The analysis provided here can be seen as the best the media can achieve over different possible communication protocols and equilibrium selection rules.

The messages produced by the source and the reports published by the media are assumed to be costless. In reality of the authoritarian states, messages that suggest the ruler's incompetence may be associated with the consequential punishment. The introduction of cost associated with specific messages imposes modeling challenges and makes the methodology developed in this paper futile. Instead, one would need to make use of, for example, the methodology of the papers that study strategic communication with lying costs as in [Kartik \(2009\)](#).

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Appendix

Proof of Lemma 1.

The proof is standard and thence omitted (see, for example, [Myerson \(1982\)](#)).

Proof of Lemma 2.

The honesty constraint (2) can be rewritten as follows: for every $\theta, \theta' \in \Theta$,

$$(u_S(a_1, \theta) - u_S(a_0, \theta)) (\pi(a_1|\theta) - \pi(a_1|\theta')) \geq 0.$$

Pick two source types θ' and θ'' that prefer a_1 over a_0 , that is, $\theta', \theta'' \in \Theta_1$. Then from the inequality above, $\pi(a_1|\theta') \geq \pi(a_1|\theta'')$ and $\pi(a_1|\theta'') \geq \pi(a_1|\theta')$, and consequently $\pi(a_1|\theta') = \pi(a_1|\theta'')$. Since θ' and θ'' were chosen arbitrarily from Θ_1 , $\pi(a_1|\theta)$ is constant across $\theta \in \Theta_1$. Call this constant π_1 . Similarly, $\pi(a_1|\theta)$ is constant across $\theta \in \Theta_0$. Call this constant π_0 . Thus, the honesty constraint is equivalent to the following condition: for every $\theta_0 \in \Theta_0$ and $\theta_1 \in \Theta_1$,

$$(u_S(a_1, \theta_0) - u_S(a_0, \theta_0)) (\pi_0 - \pi_1) \geq 0,$$

$$(u_S(a_1, \theta_1) - u_S(a_0, \theta_1)) (\pi_1 - \pi_0) \geq 0.$$

Both of these inequalities are equivalent to $\pi_1 \geq \pi_0$.

Proof of Proposition 1.

By Lemma 2, the obedience constraint (3) can be written as follows:

$$\pi_0 \int_{\Theta_0} \delta_R(\theta) d\mu_0 + \pi_1 \int_{\Theta_1} \delta_R(\theta) d\mu_0 \geq 0,$$

or $\pi_0 I_0 + \pi_1 I_1 \geq 0$. There are three cases to consider, depending on the signs of I_0 and I_1 .

The case in which $I_0 \geq 0$ and $I_1 \geq 0$ is ruled out by the tension condition (1).

Case 1. If $I_0 < 0$ and $I_1 < 0$, then the only way to satisfy the obedience constraint is to set

$$\pi_0 = \pi_1 = 0.$$

Case 2. If $I_0 \geq 0$ and $I_1 < 0$, then $\pi_1 \leq \frac{I_0}{-I_1} \cdot \pi_0$. By Lemma 2, $\pi_1 \geq \pi_0$. Finally, $\frac{I_0}{-I_1} < 1$, since by the tension condition (1), $I_0 + I_1 < 0$. The only way to satisfy these inequalities is again to set $\pi_0 = \pi_1 = 0$.

Case 3. Suppose $I_0 < 0$ and $I_1 \geq 0$. If $I_1 = 0$, then in the honest and obedient protocol, $\pi_0 = 0$ and $\pi_1 \in [0, 1]$. If $I_1 > 0$, then

$$\pi_1 \geq \frac{-I_0}{I_1} \cdot \pi_0 \geq \pi_0,$$

where the second inequality is implied by the tension condition (1). Thus, the set of honest and obedient protocols for this case is the set of $\pi_0, \pi_1 \in [0, 1]$, such that $\pi_1 \geq \frac{-I_0}{I_1} \cdot \pi_0$.

Proof of Proposition 2.

If $I_1 < 0$, then the only incentive-compatible protocol is $\pi_0 = \pi_1 = 0$. The associated media's payoff is then 0.

If $I_1 = 0$, then $\pi_0 = 0$. The media choose π_1 as high as possible, that is, $\pi_1 = 1$.

Finally, if $I_1 > 0$, then the set of incentive-compatible protocols is the triangle depicted in Figure 1. According to the media's objective, the media want the pair (π_0, π_1) to be as high as possible. The solution then is the extreme point $\left(\frac{I_1}{-I_0}, 1\right)$. The corresponding payoff is then

$$\begin{aligned} \mu_0(\Theta_0) \cdot \frac{I_1}{-I_0} + \mu_0(\Theta_1) &= \mu_0(\Theta_0) \cdot \frac{\int_{\Theta_1} \delta_R(\theta) d\mu_0}{-\int_{\Theta_0} \delta_R(\theta) d\mu_0} + \mu_0(\Theta_1) = \\ &= \mu_0(\Theta_1) \cdot \frac{\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_1] - \mathbb{E}[\delta_R(\theta)|\theta \in \Theta_0]}{-\mathbb{E}[\delta_R(\theta)|\theta \in \Theta_0]}. \end{aligned}$$

Proof of Claims 1, 2, 3.

These results directly follow from Proposition 2.

Proof of Claim 4.

The media's problem facing no honesty constraints can be written as follows:

$$\max_{\pi(a_1|\theta) \in [0,1]^{[0,1]}} \int_0^1 \pi(a_1|\theta) d\theta,$$

subject to the obedience constraint (3) tailored to the example:

$$\int_0^1 (\theta - \omega) \pi(a_1|\theta) d\theta \geq 0.$$

First, note that $\pi(a_1|\theta) = 1$ for $\theta \geq \omega$. Indeed, this choice relaxes the obedience constraint as much as possible and provides the maximal payoff to the media for $\theta \geq \omega$. Thus, the

problem is reduced to

$$\max_{\pi(a_1|\theta) \in [0,1]^{[0,\omega]}} \int_0^\omega \pi(a_1|\theta) d\theta + 1 - \omega,$$

subject to

$$\int_0^\omega (\omega - \theta) \pi(a_1|\theta) d\theta \leq \frac{(1 - \omega)^2}{2}.$$

But then the solution is $\pi(a_1|\theta) = 1$ for $2\omega - 1 \leq \theta < \omega$, since those types are associated with “cheaper” cost of persuasion, namely, $\omega - \theta$. The optimal information structure then follows. The media’s payoff is the length of the interval $[2\omega - 1, 1]$, which is $2 - 2\omega$.

Proof of Lemma 3.

The argument is standard and can be found in Proposition 3 in [Bergemann and Morris \(2019\)](#).

Proof of Lemma 4.

Suppose $\Omega_1 \in \text{supp}\phi$. Let $b = \sup(\Omega_1) \in [0, 1]$. Note that the left-hand side of inequalities (6) and (7) is continuous and strictly decreasing in ω , as $\delta_R(\theta, \omega) = \theta - \omega$. Hence, any $\omega < b$ has to lie in Ω_1 by (6). Similarly, any $\omega > b$ has to lie in $[0, 1] \setminus \Omega_1$ by (7). Thus, $\Omega_1 = [0, b]$. Finally, since δ is continuous in ω ,

$$\int_0^1 \delta_R(\cdot, b) d\phi([0, b]|\cdot) = 0$$

has to be satisfied.

Proof of Lemma 5.

Using Lemma 4, the honesty constraint (5) can be written as follows:

$$(u_S(a_1, \theta) - u_S(a_0, \theta)) \sum_{[0,b] \in \text{supp}\phi} (\phi([0, b]|\theta) - \phi([0, b]|\theta')) H(b) \geq 0$$

for every $\theta, \theta' \in \Theta$. Thus, by the same argument as in Lemma 2, for every $\theta', \theta'' \in \Theta_1$, $\sum_{[0,b] \in \text{supp}\phi} \phi([0, b]|\theta') H(b) = \sum_{[0,b] \in \text{supp}\phi} \phi([0, b]|\theta'') H(b)$. Then there exists $s_1 \in [0, 1]$, such that $s_1 = \sum_{[0,b] \in \text{supp}\phi} \phi([0, b]|\theta) H(b)$ for every $\theta \in \Theta_1$. The same argument shows that $s_0 = \sum_{[0,b] \in \text{supp}\phi} \phi([0, b]|\theta) H(b)$ for every $\theta \in \Theta_0$. Finally, the honesty constraint for some type $\theta_1 \in \Theta_1$ deliberating a misreport $\theta_0 \in \Theta_0$ pins down that $s_1 \geq s_0$.

Proof of Lemma 6.

Lemmas 3, 4, and 5 imply the constraints of the reduced problem. The specification of sets Θ_0 and Θ_1 implies the media’s objective function.

Proof of Lemma 7.

The obedience constraints pin down $\underline{b}$ and $\bar{b}$ as functions of ϕ_0 and ϕ_1 :

$$\bar{b} = f_1(\phi_0, \phi_1) = \frac{\phi_0 \int_{\Theta_0} \theta d\mu_0 + \phi_1 \int_{\Theta_1} \theta d\mu_0}{\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1)}, \quad (8)$$

$$\underline{b} = f_2(\phi_0, \phi_1) = \frac{(1 - \phi_0) \int_{\Theta_0} \theta d\mu_0 + (1 - \phi_1) \int_{\Theta_1} \theta d\mu_0}{(1 - \phi_0) \mu_0(\Theta_0) + (1 - \phi_1) \mu_0(\Theta_1)}. \quad (9)$$

Then the boundary conditions are established. If $\phi_0 = \phi_1$, then $\underline{b} = \bar{b} = \mathbb{E}[\theta]$. If $\phi_0 = 0$, then $\bar{b} = \mathbb{E}[\theta | \theta \in \Theta_1]$. If $\phi_1 = 1$, then $\underline{b} = \mathbb{E}[\theta | \theta \in \Theta_0]$. The derivatives of $\bar{b}$ and $\underline{b}$ can be directly calculated:

$$\begin{aligned} \frac{\partial \bar{b}}{\partial \phi_0} &= \frac{\int_{\Theta_0} \theta d\mu_0 \cdot (\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1)) - (\phi_0 \int_{\Theta_0} \theta d\mu_0 + \phi_1 \int_{\Theta_1} \theta d\mu_0) \cdot \mu_0(\Theta_0)}{(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1))^2} \\ &= \frac{\phi_1 \mu_0(\Theta_1) \int_{\Theta_0} \theta d\mu_0 - \phi_1 \mu_0(\Theta_0) \int_{\Theta_1} \theta d\mu_0}{(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1))^2} \\ &= \mu_0(\Theta_0) \mu_0(\Theta_1) \cdot \frac{\phi_1 (\mathbb{E}[\theta | \theta \in \Theta_0] - \mathbb{E}[\theta | \theta \in \Theta_1])}{(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1))^2} < 0. \\ \frac{\partial \bar{b}}{\partial \phi_1} &= \frac{\int_{\Theta_1} \theta d\mu_0 \cdot (\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1)) - (\phi_0 \int_{\Theta_0} \theta d\mu_0 + \phi_1 \int_{\Theta_1} \theta d\mu_0) \cdot \mu_0(\Theta_1)}{(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1))^2} \\ &= \frac{\phi_0 \mu_0(\Theta_0) \int_{\Theta_1} \theta d\mu_0 - \phi_0 \mu_0(\Theta_1) \int_{\Theta_0} \theta d\mu_0}{(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1))^2} \\ &= \mu_0(\Theta_0) \mu_0(\Theta_1) \cdot \frac{\phi_0 (\mathbb{E}[\theta | \theta \in \Theta_1] - \mathbb{E}[\theta | \theta \in \Theta_0])}{(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1))^2} > 0. \end{aligned}$$

Similarly, it can be shown that $\frac{\partial \underline{b}}{\partial \phi_0} > 0$ and $\frac{\partial \underline{b}}{\partial \phi_1} < 0$. The bounds on $\bar{b}$ and $\underline{b}$ in the statement of the lemma are then implied.

The signs of these derivatives are intuitive. For example, if ϕ_0 increases, then the probability of getting the recommendation $[0, \bar{b}]$ goes up for the lower competence levels $\theta \in \Theta_0 = [0, \bar{\theta}]$. Thus, some receivers with high ω will find it optimal to switch from the mobilizing action to the status-quo action. That is, the obedience constraint will make $\bar{b}$ lower.

Given the boundary conditions and the signs of the derivatives above, there always exist $\phi_0, \phi_1 \in [0, 1]$, with $\phi_1 \geq \phi_0$, such that $\bar{b} = f_1(\phi_0, \phi_1)$ and $\underline{b} = f_2(\phi_0, \phi_1)$ by a multivariate version of the mean value theorem.

It is worth mentioning that under our assumptions on μ_0 , Θ_0 , and Θ_1 , the system of equations $\bar{b} = f_1(\phi_0, \phi_1)$, $\underline{b} = f_2(\phi_0, \phi_1)$ can be solved directly. It is a matter of algebra to show that

$$\phi_0 = \frac{1}{\bar{\theta}} \cdot \frac{(1 - 2\bar{b} + \bar{\theta}) \left(\frac{1}{2} - \underline{b}\right)}{\bar{b} - \underline{b}},$$

and

$$\phi_1 = \frac{2}{1 - \bar{\theta}} \cdot \frac{\left(\bar{b} - \frac{\bar{\theta}}{2}\right) \left(\frac{1}{2} - \underline{b}\right)}{\bar{b} - \underline{b}},$$

if $\bar{b} > \underline{b}$. These ϕ_0 and ϕ_1 can be readily checked to satisfy $\phi_0, \phi_1 \in [0, 1]$ and $\phi_1 \geq \phi_0$. If $\bar{b} = \underline{b}$, then $\bar{b} = \underline{b} = \mathbb{E}[\theta]$ which occurs as long as $\phi_0 = \phi_1$.

Proof of Proposition 3.

Equations (8) and (9) can be combined to get

$$\underline{b} + (\bar{b} - \underline{b})(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1)) = \mathbb{E}[\theta].$$

By Lemma 6, the objective of the media is

$$\begin{aligned} \mu_0(\Theta_0)s_0 + \mu_0(\Theta_1)s_1 &= H(\underline{b}) + (H(\bar{b}) - H(\underline{b}))(\phi_0 \mu_0(\Theta_0) + \phi_1 \mu_0(\Theta_1)) \\ &= H(\underline{b}) + \frac{\mathbb{E}[\theta] - \underline{b}}{\bar{b} - \underline{b}}(H(\bar{b}) - H(\underline{b})) = \frac{\bar{b} - \mathbb{E}[\theta]}{\bar{b} - \underline{b}}H(\underline{b}) + \frac{\mathbb{E}[\theta] - \underline{b}}{\bar{b} - \underline{b}}H(\bar{b}), \end{aligned}$$

as long as $\bar{b} > \underline{b}$. If $\bar{b} = \underline{b}$, then the objective of the media is $H(\mathbb{E}[\theta])$. Note that

$$\frac{\bar{b} - \mathbb{E}[\theta]}{\bar{b} - \underline{b}} \cdot \underline{b} + \frac{\mathbb{E}[\theta] - \underline{b}}{\bar{b} - \underline{b}} \cdot \bar{b} = \mathbb{E}[\theta].$$

By Lemma 7, any $\bar{b} \in [\mathbb{E}[\theta], \mathbb{E}[\theta|\theta \in \Theta_1]]$ and $\underline{b} \in [\mathbb{E}[\theta|\theta \in \Theta_0], \mathbb{E}[\theta]]$ can be achieved by some simple protocol. Therefore, the media's problem is a splitting problem with the value function $\text{cav} \hat{H}[E[\theta]]$ (Le Treust and Tomala, 2019). The corresponding $\bar{b}$ and $\underline{b}$ can then be established as the supporting points of this object.